

ARAK, Iran, Islamic Republic of

WMO: 407690

Lat: 34.0716N

Lon: 49.7837E

Elev: 5587

StdP: 11.96

Time Zone: 3.50 (IRN)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.3	15.1	0.4	6.9	20.8	5.5	9.0	33.2	22.7	44.3	19.7	46.8	0.8	120	0.440

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.0	98.0	60.6	95.9	60.0	93.9	59.3	64.8	89.9	62.8	89.3	61.3	88.5	8.7	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
55.8	81.7	78.1	52.6	72.4	74.3	50.0	65.9	71.8	33.0	88.8	31.3	89.2	30.1	88.1	78.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.5	17.7	15.3	5.7	101.7	9.2	2.1	-0.9	103.2	-6.3	104.5	-11.4	105.7	-18.0	107.2		
(5)			4.3	67.2	9.0	3.9	-2.2	70.0	-7.4	72.2	-12.4	74.4	-18.9	77.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	57.9	32.4	38.0	47.1	56.0	65.9	76.6	82.1	80.2	71.6	60.3	46.0	37.6		
	DBStd	18.01	10.42	8.84	7.43	6.12	5.13	4.52	3.70	3.16	4.13	5.68	6.65	6.89		
	HDD50	1587	545	340	142	19	0	0	0	0	0	2	153	386		
	HDD65	4236	1010	756	555	275	52	1	0	0	4	165	570	848		
	CDD50	4478	0	4	53	198	493	798	994	935	647	321	33	2		
	CDD65	1653	0	0	2	4	80	349	529	470	201	18	0	0		
	CDH74	21012	0	0	4	47	875	4248	7044	6041	2460	293	0	0		
	CDH80	10456	0	0	0	2	191	2044	3968	3269	955	27	0	0		
	WSAvg	4.5	3.1	4.5	5.8	5.8	5.7	5.4	5.3	4.5	4.2	4.0	3.0	2.6		
	PrecAvg	10.5	1.2	1.2	1.9	1.7	0.8	0.1	0.1	0.1	0.0	0.6	1.3	1.4		
(15) Precipitation	PrecMax	15.5	2.6	3.2	4.1	4.1	3.1	0.8	0.4	1.0	0.7	2.2	5.0	4.8		
	PrecMin	6.3	0.0	0.1	0.0	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	2.4	0.7	0.7	1.1	1.2	0.8	0.2	0.1	0.2	0.2	0.6	1.1	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.6	61.9	73.8	79.2	87.7	97.8	101.3	99.1	92.7	82.8	69.6	61.9		
		MCWB	40.8	44.2	49.5	52.8	56.5	59.5	61.3	61.7	57.8	53.5	48.2	42.8		
	2%	DB	52.7	57.3	68.8	74.5	84.6	94.9	99.1	96.9	90.5	80.1	65.5	56.4		
		MCWB	39.6	42.2	46.5	51.4	55.3	58.6	60.7	60.3	57.2	52.8	46.0	42.0		
	5%	DB	49.3	54.1	64.5	71.6	81.9	92.5	97.0	95.1	88.3	77.4	62.0	52.3		
		MCWB	38.6	40.6	45.1	50.2	54.3	58.1	60.5	59.4	56.1	51.5	45.6	40.6		
	10%	DB	46.0	50.9	60.7	68.5	79.1	90.0	94.9	93.1	85.8	74.3	58.3	48.7		
		MCWB	36.8	39.3	43.6	49.3	53.7	57.4	60.2	59.0	55.2	50.5	44.5	39.1		
	0.4%	WB	43.6	46.4	51.6	55.5	60.0	65.2	69.0	68.3	62.3	56.7	51.2	45.9		
		MCDB	51.4	56.7	68.8	71.1	79.5	88.0	89.7	93.6	87.2	76.4	62.2	55.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	41.4	43.9	48.4	53.7	58.2	62.3	65.6	63.9	59.3	54.5	49.1	43.5		
		MCDB	49.4	54.6	63.1	69.9	78.4	86.7	89.6	91.2	85.8	74.0	60.2	52.6		
	5%	WB	39.4	41.9	46.4	52.2	56.8	60.5	63.4	62.0	57.6	53.0	47.2	41.7		
		MCDB	47.4	51.7	61.0	67.5	76.3	86.4	89.8	89.3	83.8	72.2	57.9	50.2		
	10%	WB	37.4	40.1	44.8	50.8	55.4	59.1	61.9	60.6	56.3	51.7	45.6	40.0		
		MCDB	45.3	49.4	58.0	64.9	74.0	85.6	89.4	88.6	82.1	70.0	55.6	47.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.7	18.4	20.3	20.4	22.7	25.2	25.0	26.7	28.7	25.3	19.8	17.3		
		MCDBR	21.4	22.8	24.7	24.4	26.1	27.8	27.4	29.0	30.9	28.8	25.2	22.8		
		MCWBR	12.0	11.7	10.7	9.1	8.9	10.0	9.3	10.3	11.8	12.1	11.8	12.8		
		MCDBR	19.5	20.5	21.5	21.5	23.6	25.2	24.4	26.6	28.7	26.2	21.0	19.9		
	5% WB	MCWBR	11.7	11.6	10.5	9.3	9.2	10.2	10.0	10.7	11.7	11.9	10.6	12.0		
(40) Clear-Sky Solar Irradiance	taub		0.287	0.324	0.380	0.469	0.463	0.449	0.489	0.438	0.374	0.381	0.326	0.287		
	taud		2.414	2.304	2.123	1.907	1.946	1.993	1.901	2.033	2.206	2.157	2.337	2.453		
	Ebn at Noon		294	292	282	261	264	266	255	266	280	266	275	288		
	Edh at Noon		29	36	47	61	60	57	62	53	43	41	31	26		
(44) All-Sky Solar Radiation	RadAvg		933	1222	1586	1842	2200	2598	2459	2310	2000	1470	1022	840		
	RadStd		76	102	128	152	155	101	93	78	72	92	93	100		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	N/A	N/A	N/A	+1.30	-1.53	-2.18	N/A	N/A	N/A	N/A

Nomenclature: See separate page